

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the function $h(x) = \int_0^x t^2 dt$.

- a) Fill out the following table of values for the function $h(x)$

x	-3	-1	0	1	3
$h(x)$					

- b) Provide a rough sketch of what the function $h(x)$ looks like.
 c) Determine the value of $h'(2)$.

2. Find the derivative of the function $g(x) = \int_x^8 [e^t - \ln(t^2 + 1)] dt$.

3. Find the derivative of the function $g(x) = \int_4^{\ln(x^2+1)} \left[\frac{\sin(t) + 2}{e^t + t} \right] dt$.

4. Compute $\frac{d}{ds} \left[\int_{\cos(s)}^{s^2+2} \frac{1}{x^2+1} dx \right]$

5. Let $g(x) = \int_x^{x^2} 6 dt$.

- a) Use your calculator to graph this function on the interval $[-1, 1]$.
 b) Compute $g'(x)$ by first evaluating the integral using the FTC2 and then differentiating.
 c) Compute $g'(x)$ by using the FTC1 directly. Note that you will have to rewrite the function a bit first.
 d) Show that the two results from parts (b) and (c) are consistent.

6. Determine the equation of the tangent line to $g(x) = \int_0^{x^2+x} t \, dt$ at $x = 3$.

7. Let $F(x) = \int_0^x \frac{t-3}{t^2+7} \, dt$.

a) Find all intervals where $F(x)$ is increasing.

b) Find all intervals where $F(x)$ is concave up.

8. Compute the maximum of the function $g(x) = \int_{e^x}^{2e^x} \frac{1}{1+w^2} \, dw$.

9. Find the maximum value of the area of the region under the curve $y = 4x - x^3$ from $x = a$ to $x = a+1$, for all $a > 0$.

10. If $x \sin(\pi x) = \int_0^{x^2} f(t) \, dt$, where f is a continuous function, find $f(4)$.